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CIS 5250

14 Sept. 2026

Wireshark Data Capture Project

Goal: The purpose of this hands-on project is to help you practice the following skills to achieve the module learning objectives (MLOs):

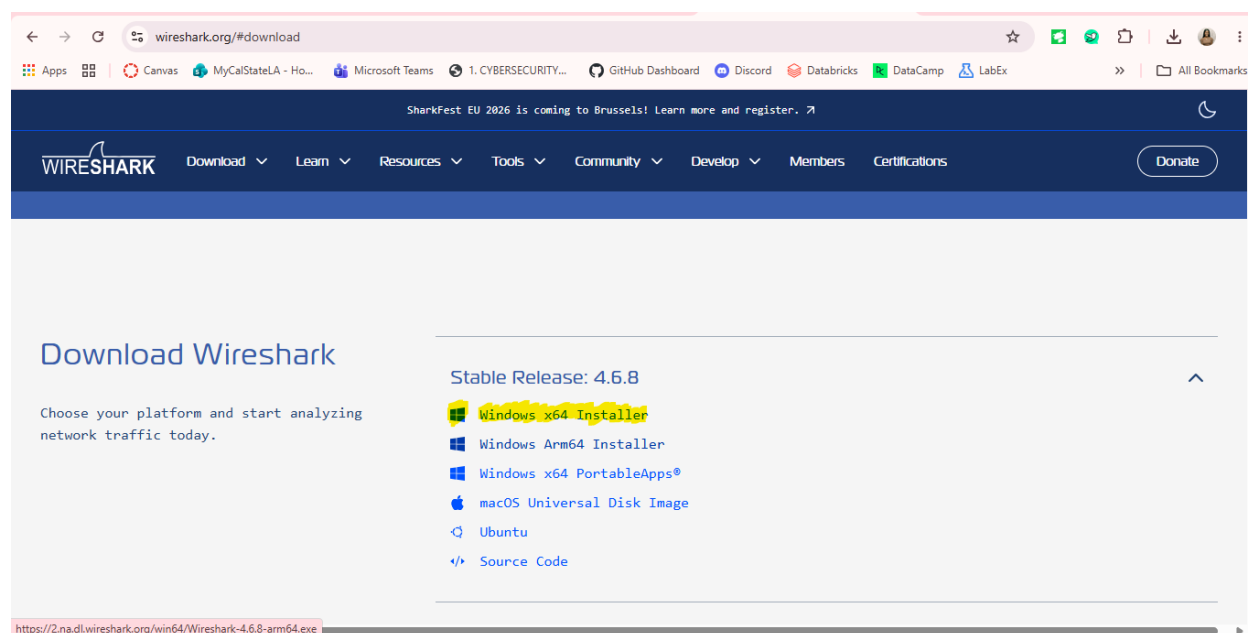
- **MLO 2.3** Analyze basic network communication between hosts using MAC addresses, IP addresses, port numbers, and client/server protocols.

Tasks

1. Study the tutorial on Capturing and Analyzing Data Packets using **Wireshark**.
2. Download and Install **Wireshark**.
3. Conduct the lab project steps of capturing data traffic using **Wireshark**.
4. Answer the listed questions.
5. Submit the project report containing the lab steps, the screenshots with appropriate descriptions, the records of your observations, and the answers to the questions.

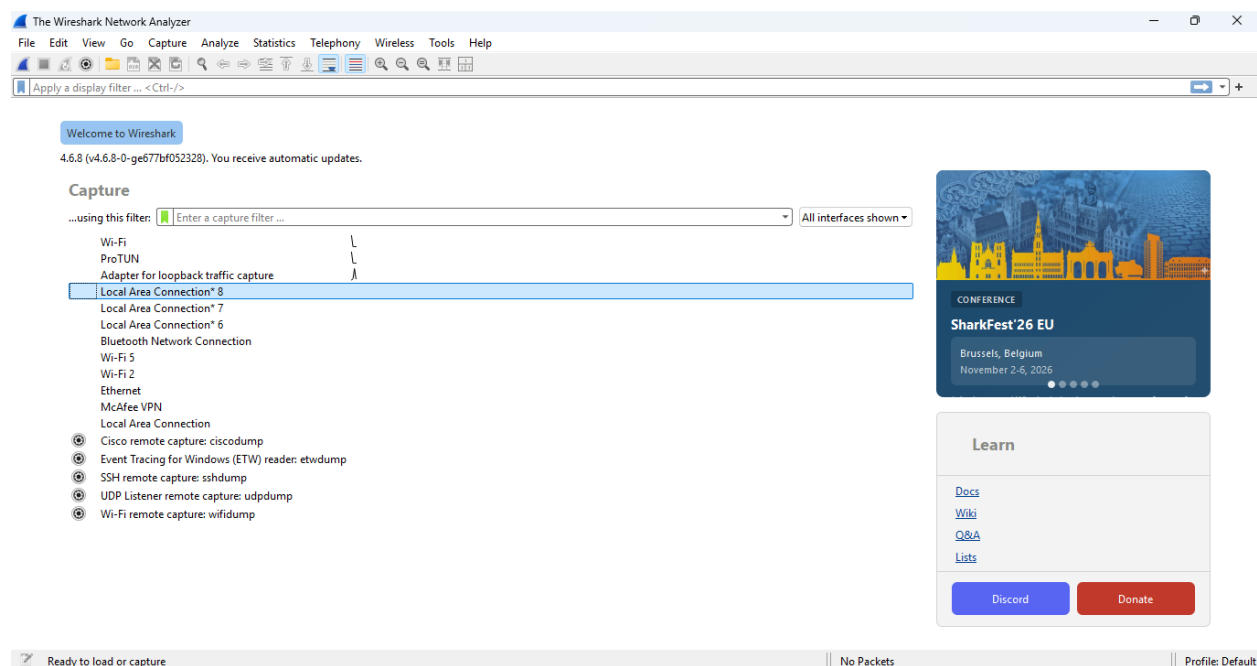
Submission

Part 1: Wireshark installation via the <http://wireshark.org/#download> website



Part 2: Capture Ping Traffic using Wireshark

Step 1: Launch Wireshark



Step 2: Open a web browser and go to www.calstatela.edu

Step 3: Open a PowerShell window. Enter the command `ipconfig /all` to find your IP address.

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\thelma> ipconfig /all

Windows IP Configuration

    Host Name . . . . . : Thelma
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Unknown adapter ProTUN:

    Connection-specific DNS Suffix . :
    Description . . . . . : Proton VPN Windows Tunnel
    Physical Address. . . . . :
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 10.2.0.2(Preferred)
    Subnet Mask . . . . . : 255.255.255.255
    Default Gateway . . . . . : 0.0.0.0
    DNS Servers . . . . . : 10.2.0.1
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . :
    Description . . . . . : VirtualBox Host-Only Ethernet Adapter

```

```

Windows PowerShell

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-18-A1-F8-A3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Unknown adapter McAfee VPN:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : TAP-Windows Adapter V9
    Physical Address. . . . . : 00-FF-66-5F-E6-52
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 2:

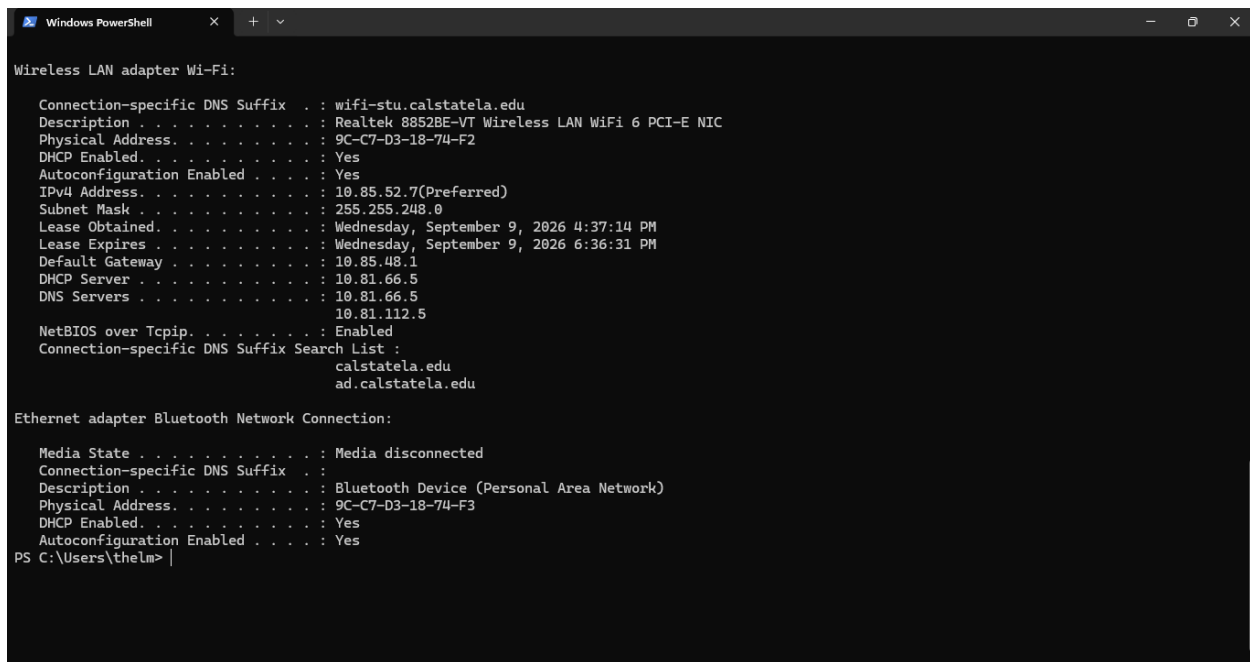
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC #2
    Physical Address. . . . . : 9E-C7-D3-18-74-F2
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 5:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC #5
    Physical Address. . . . . : 92-C7-D3-18-74-F2
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

```



```

Windows PowerShell

Wireless LAN adapter Wi-Fi:

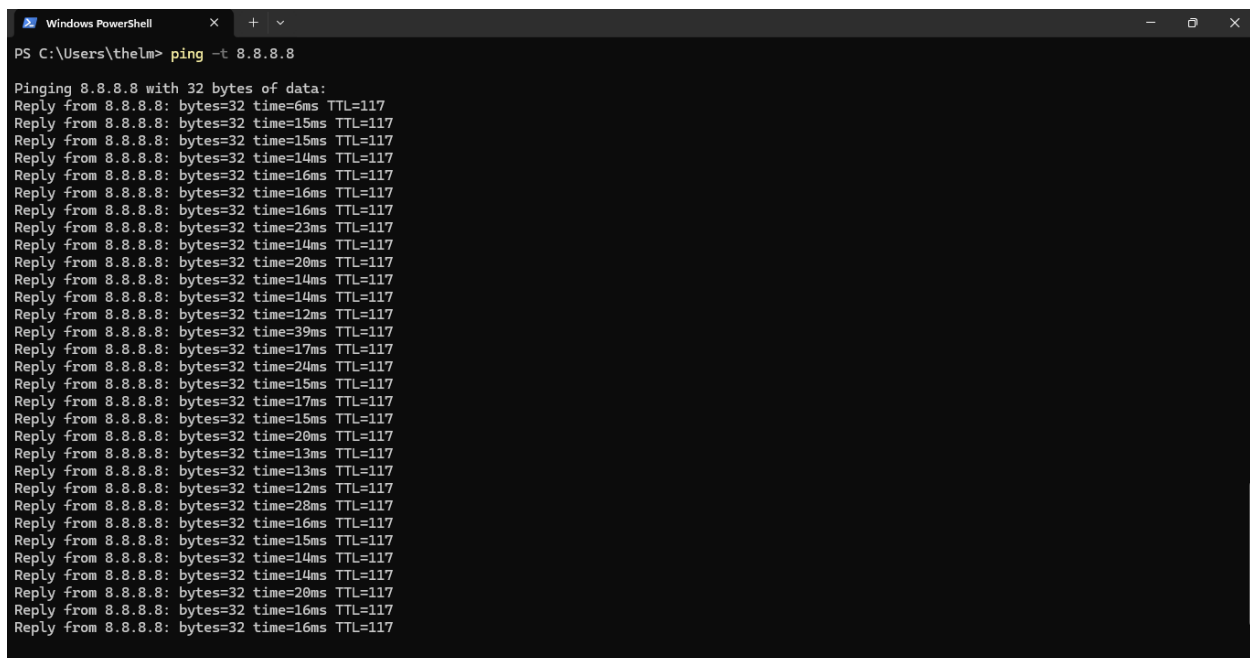
Connection-specific DNS Suffix  . : wifi-stu.calstatela.edu
Description . . . . . : Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC
Physical Address. . . . . : 9C-C7-D3-18-74-F2
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 10.85.52.7(Preferred)
Subnet Mask . . . . . : 255.255.248.0
Lease Obtained. . . . . : Wednesday, September 9, 2026 4:37:14 PM
Lease Expires . . . . . : Wednesday, September 9, 2026 6:36:31 PM
Default Gateway . . . . . : 10.85.48.1
DHCP Server . . . . . : 10.81.66.5
DNS Servers . . . . . : 10.81.66.5
                          10.81.112.5
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
                          calstatela.edu
                          ad.calstatela.edu

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 9C-C7-D3-18-74-F3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
PS C:\Users\thelma>

```

Step 4: Enter `ping -t 8.8.8.8` and press Enter. The computer begins pinging Google's DNS server and will continue until you stop it.



```

Windows PowerShell

PS C:\Users\thelma> ping -t 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=6ms TTL=117
Reply from 8.8.8.8: bytes=32 time=15ms TTL=117
Reply from 8.8.8.8: bytes=32 time=15ms TTL=117
Reply from 8.8.8.8: bytes=32 time=14ms TTL=117
Reply from 8.8.8.8: bytes=32 time=16ms TTL=117
Reply from 8.8.8.8: bytes=32 time=16ms TTL=117
Reply from 8.8.8.8: bytes=32 time=16ms TTL=117
Reply from 8.8.8.8: bytes=32 time=23ms TTL=117
Reply from 8.8.8.8: bytes=32 time=14ms TTL=117
Reply from 8.8.8.8: bytes=32 time=20ms TTL=117
Reply from 8.8.8.8: bytes=32 time=14ms TTL=117
Reply from 8.8.8.8: bytes=32 time=14ms TTL=117
Reply from 8.8.8.8: bytes=32 time=12ms TTL=117
Reply from 8.8.8.8: bytes=32 time=39ms TTL=117
Reply from 8.8.8.8: bytes=32 time=17ms TTL=117
Reply from 8.8.8.8: bytes=32 time=24ms TTL=117
Reply from 8.8.8.8: bytes=32 time=15ms TTL=117
Reply from 8.8.8.8: bytes=32 time=17ms TTL=117
Reply from 8.8.8.8: bytes=32 time=15ms TTL=117
Reply from 8.8.8.8: bytes=32 time=20ms TTL=117
Reply from 8.8.8.8: bytes=32 time=13ms TTL=117
Reply from 8.8.8.8: bytes=32 time=13ms TTL=117
Reply from 8.8.8.8: bytes=32 time=12ms TTL=117
Reply from 8.8.8.8: bytes=32 time=28ms TTL=117
Reply from 8.8.8.8: bytes=32 time=16ms TTL=117
Reply from 8.8.8.8: bytes=32 time=15ms TTL=117
Reply from 8.8.8.8: bytes=32 time=14ms TTL=117
Reply from 8.8.8.8: bytes=32 time=14ms TTL=117
Reply from 8.8.8.8: bytes=32 time=20ms TTL=117
Reply from 8.8.8.8: bytes=32 time=16ms TTL=117
Reply from 8.8.8.8: bytes=32 time=16ms TTL=117

```

Step 5: When the total number of ping responses in PowerShell is greater than 10, in the Wireshark window, click Capture and then click the red box Stop on the command ribbon to stop the capture. Return to the PowerShell window and press CTRL+C to stop

the ping

Wireshark capture showing network traffic. The packet list displays 15 packets, all of type 'Transport Data' from WireGuard. The packet details pane shows the structure of a WireGuard packet: Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and WireGuard Protocol.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.85.52.7	146.70.230.98	WireGu...	134	Transport Data, receiver=0x52DC1060, counter=174, datalen=60
2	0.007116300	146.70.230.98	10.85.52.7	WireGu...	154	Transport Data, receiver=0xC799DD0A, counter=196, datalen=80
3	0.008266000	10.85.52.7	146.70.230.98	WireGu...	150	Transport Data, receiver=0x52DC1060, counter=175, datalen=76
4	0.008969200	146.70.230.98	10.85.52.7	WireGu...	138	Transport Data, receiver=0xC799DD0A, counter=197, datalen=64
5	0.014628200	146.70.230.98	10.85.52.7	WireGu...	122	Transport Data, receiver=0xC799DD0A, counter=198, datalen=48
6	1.026779100	10.85.52.7	146.70.230.98	WireGu...	134	Transport Data, receiver=0x52DC1060, counter=176, datalen=60
7	1.037367100	146.70.230.98	10.85.52.7	WireGu...	138	Transport Data, receiver=0xC799DD0A, counter=199, datalen=64
8	2.040094300	10.85.52.7	146.70.230.98	WireGu...	134	Transport Data, receiver=0x52DC1060, counter=177, datalen=60
9	2.053426400	146.70.230.98	10.85.52.7	WireGu...	138	Transport Data, receiver=0xC799DD0A, counter=200, datalen=64
10	2.544406500	146.70.230.98	10.85.52.7	WireGu...	138	Transport Data, receiver=0xC799DD0A, counter=201, datalen=64
11	2.546298600	10.85.52.7	146.70.230.98	WireGu...	142	Transport Data, receiver=0x52DC1060, counter=178, datalen=68
12	2.550070400	146.70.230.98	10.85.52.7	WireGu...	122	Transport Data, receiver=0xC799DD0A, counter=202, datalen=48
13	3.053723000	10.85.52.7	146.70.230.98	WireGu...	134	Transport Data, receiver=0x52DC1060, counter=179, datalen=60
14	3.063754600	146.70.230.98	10.85.52.7	WireGu...	138	Transport Data, receiver=0xC799DD0A, counter=203, datalen=64
15	4.064096300	10.85.52.7	146.70.230.98	WireGu...	134	Transport Data, receiver=0x52DC1060, counter=180, datalen=60

Frame 1: Packet, 134 bytes on wire (1072 bits), 134 bytes captured (1072 bits) on interface \Device\NPF...
 Ethernet II, Src: AzureWaveTec_18:74:f2 (9c:c7:d3:18:74:f2), Dst: AlcatelLucen_bf:3e:45 (94:24:e1:bf:3e:45)
 Internet Protocol Version 4, Src: 10.85.52.7, Dst: 146.70.230.98
 User Datagram Protocol, Src Port: 52269, Dst Port: 443
 WireGuard Protocol

wireshark_Wi-FiLL9KV3.pcapng | Packets: 46 · Dropped: 0 (0.0%) | Profile: Default

##comment – I noticed all the data packets were using a ‘wireguard’ protocol and after some research, I found that this is a modern VPN protocol. It basically creates a secure tunnel between your device and a VPN server. I then turned off my VPN and tried to capture again. The result is below:

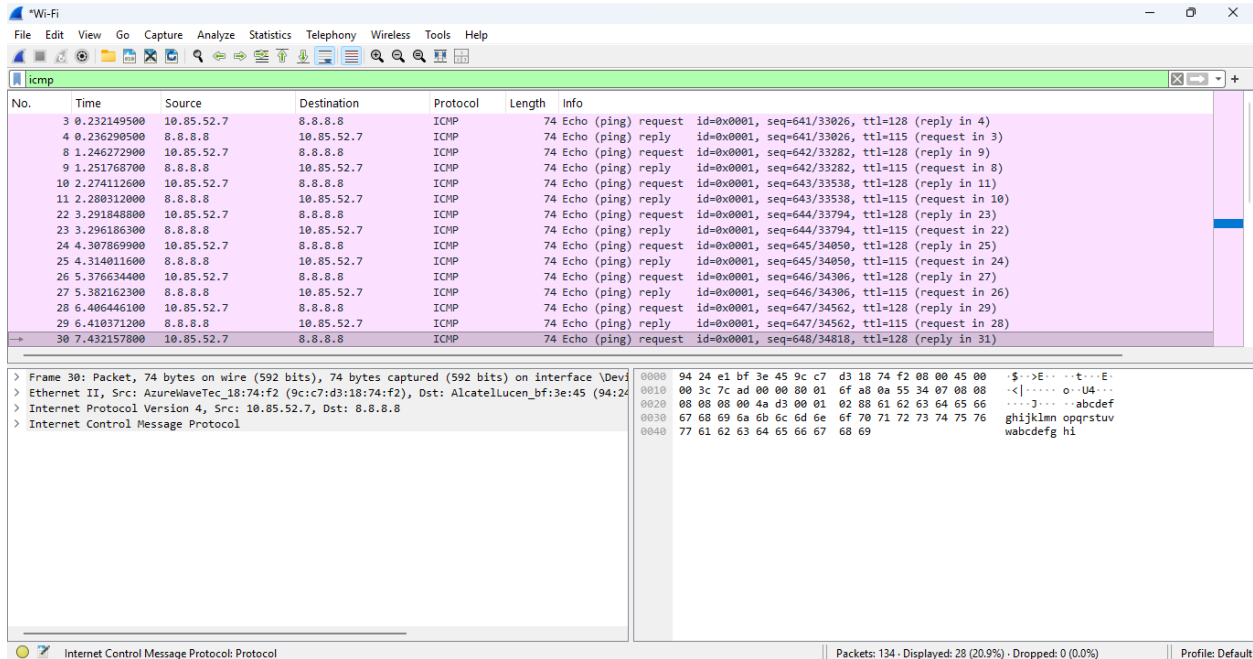
Wireshark capture showing network traffic after turning off the VPN. The packet list displays 15 packets, including ARP, ICMP, TCP, and TLS. The packet details pane shows the structure of an ARP Announcement packet: Ethernet II, Address Resolution Protocol (ARP Announcement).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	AlcatelLucen_bf:3e:45	Broadcast	ARP	60	ARP Announcement for 10.85.40.1
2	0.000256900	AlcatelLucen_bf:3e:45	Broadcast	ARP	60	ARP Announcement for 10.85.168.1
3	0.232149500	10.85.52.7	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=641/33026, ttl=128 (reply in 4)
4	0.236200500	8.8.8.8	10.85.52.7	ICMP	74	Echo (ping) reply id=0x0001, seq=641/33026, ttl=115 (request in 3)
5	0.827124500	74.179.65.112	10.85.52.7	TLSv1.2	113	Application Data
6	0.830285900	10.85.52.7	74.179.65.112	TLSv1.2	120	Application Data
7	0.863717600	74.179.65.112	10.85.52.7	TCP	60	443 → 51037 [ACK] Seq=60 Ack=67 Win=303 Len=0
8	1.246272900	10.85.52.7	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=642/33282, ttl=128 (reply in 9)
9	1.251768700	8.8.8.8	10.85.52.7	ICMP	74	Echo (ping) reply id=0x0001, seq=642/33282, ttl=115 (request in 8)
10	1.274112600	10.85.52.7	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=643/33538, ttl=128 (reply in 11)
11	2.280312000	8.8.8.8	10.85.52.7	ICMP	74	Echo (ping) reply id=0x0001, seq=643/33538, ttl=115 (request in 10)
12	2.285454400	104.18.23.222	10.85.52.7	TLSv1.2	86	Application Data
13	2.286039800	10.85.52.7	104.18.23.222	TLSv1.2	90	Application Data
14	2.300303600	104.18.23.222	10.85.52.7	TCP	60	443 → 57574 [ACK] Seq=33 Ack=37 Win=16 Len=0
15	2.353808600	10.85.52.7	20.59.87.227	TLSv1.2	153	Application Data

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF...
 Ethernet II, Src: AlcatelLucen_bf:3e:45 (94:24:e1:bf:3e:45), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 Address Resolution Protocol (ARP Announcement)

wireshark_Wi-FiR05HV3.pcapng | Packets: 134 · Dropped: 0 (0.0%) | Profile: Default

Step 7: To filter the captured frames to those involving ICMP, in the Apply a display filter field, type icmp and press Enter. The filter limits the displayed captures to ICMP messages only.



Step 8: Click a frame whose Source address is the IP address of your computer, the Protocol column shows ICMP, and the Info column shows Echo (ping) request. The frame is one of the ping requests sent to the Google DNS server. The frame is highlighted, and detailed information about the frame appears in the bottom two panes. You can also double-click the highlighted frame to display its details in a new window

Questions

1. What are the source address and the destination address in the Ethernet portion the frame? Are these addresses IP addresses or MAC addresses?

Note: This destination address in the Ethernet frame is the MAC address of the default gateway on your network.

Destination: AlcatelLucen_bf:3e:45 (94:24:e1:bf:3e:45)

Source: AzureWaveTec_18:74:f2 (9c:c7:d3:18:74:f2)

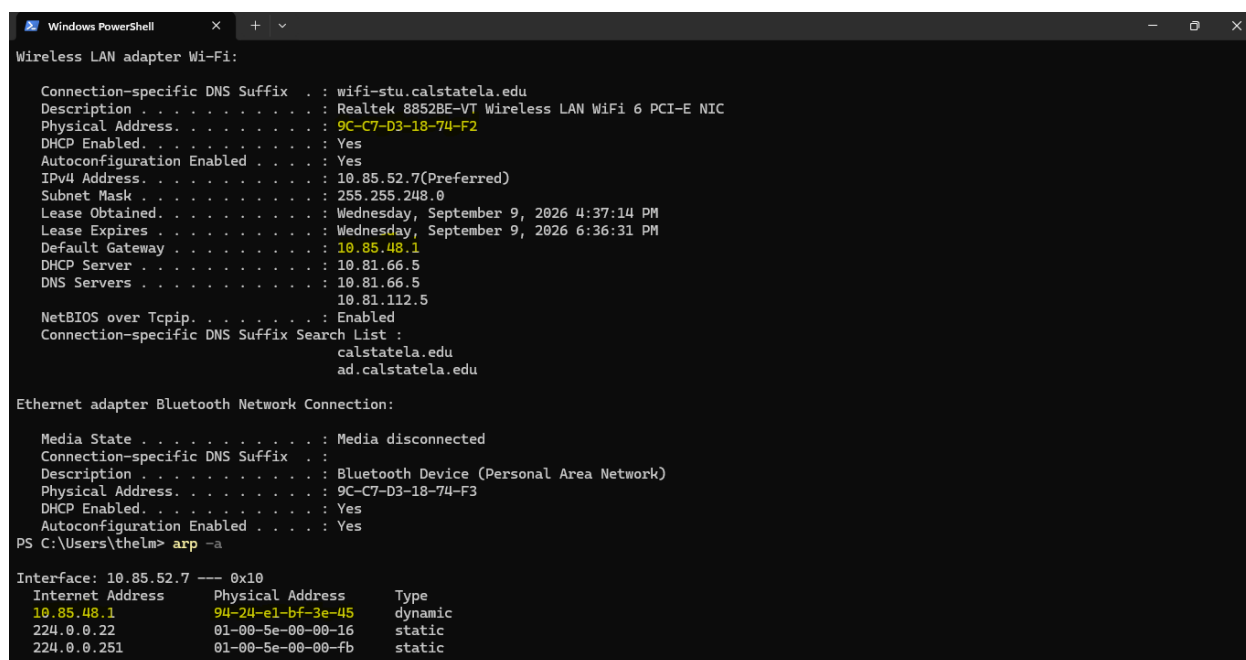
These are MAC addresses

2. The IPV4 portion of the frame is part of the payload in the Ethernet frame. Expand

this IPv4 part of the frame. **What are the destination and source IPv4 addresses in the frame?**

Internet Protocol Version 4, Src: 10.85.52.7, Dst: 8.8.8.8

Step 9: In the PowerShell window, enter the commands `ipconfig /all` and `arp -a`. Look for the MAC address of your default gateway, and your MAC and IP addresses to verify your answers in Step 8.



```
Windows PowerShell
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : wifi-stu.calstatela.edu
Description . . . . . : Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC
Physical Address. . . . . : 9C-C7-D3-18-74-F2
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 10.85.52.7(Preferred)
Subnet Mask . . . . . : 255.255.248.0
Lease Obtained. . . . . : Wednesday, September 9, 2026 4:37:14 PM
Lease Expires . . . . . : Wednesday, September 9, 2026 6:36:31 PM
Default Gateway . . . . . : 10.85.48.1
DHCP Server . . . . . : 10.81.66.5
DNS Servers . . . . . : 10.81.66.5
                        10.81.112.5
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
                        calstatela.edu
                        ad.calstatela.edu

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 9C-C7-D3-18-74-F3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

PS C:\Users\thelma> arp -a

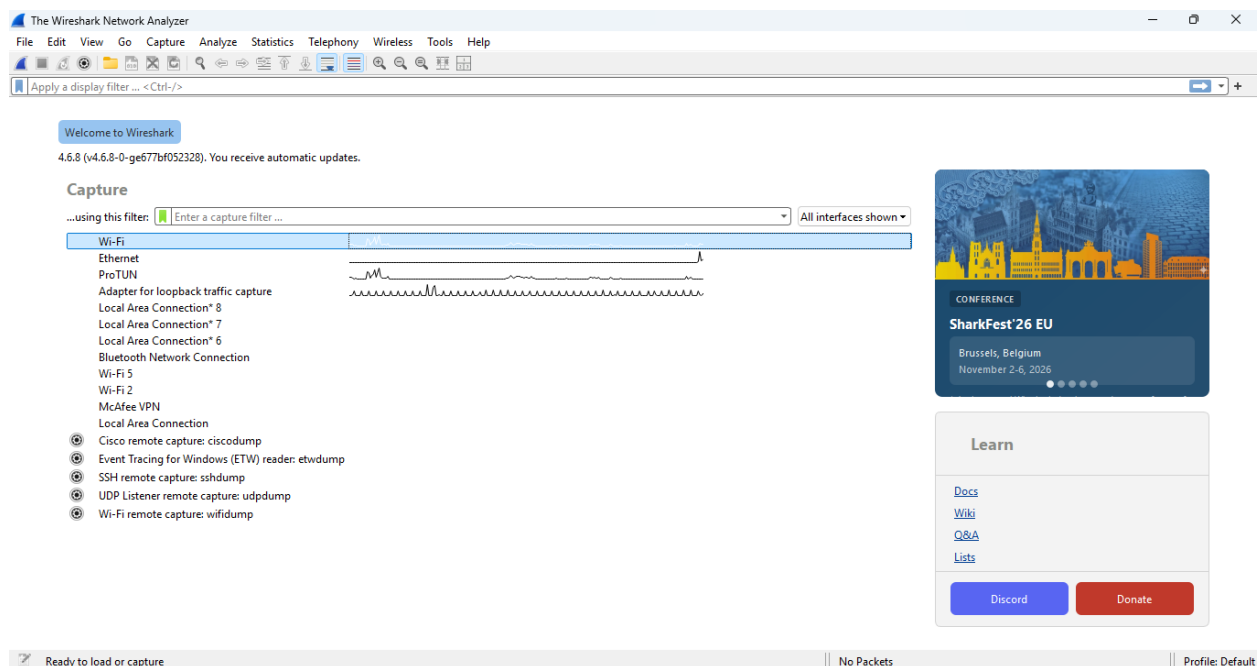
Interface: 10.85.52.7 --- 0x10
Internet Address      Physical Address      Type
10.85.48.1             94-24-e1-bf-3e-45     dynamic
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
```

##Comment - My ARP table *does confirm* the MAC address of my default gateway and it also matches the MAC address in Wireshark

Part 4: Capture Web Traffic using Wireshark

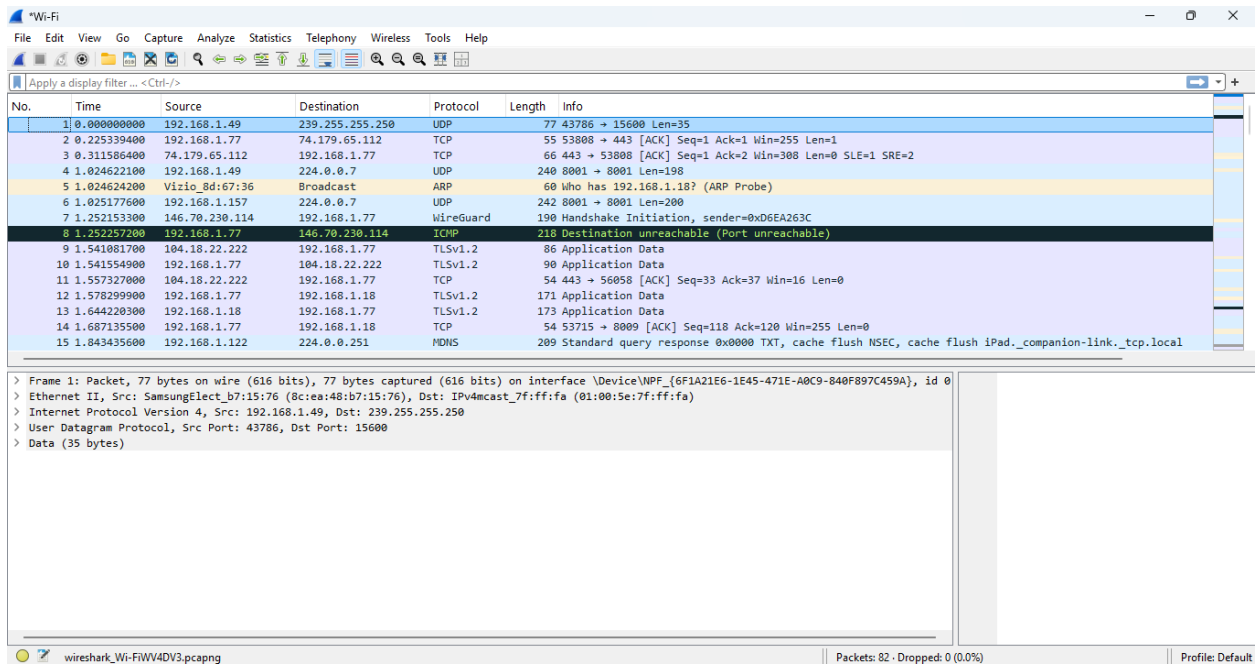
Step 1: Double click the Wireshark icon on your desktop to open it

Step 2: In the main window “Welcome to Wireshark”, in the Capture field, double-click your selected Network Interface Card (NIC) in the Interface List.



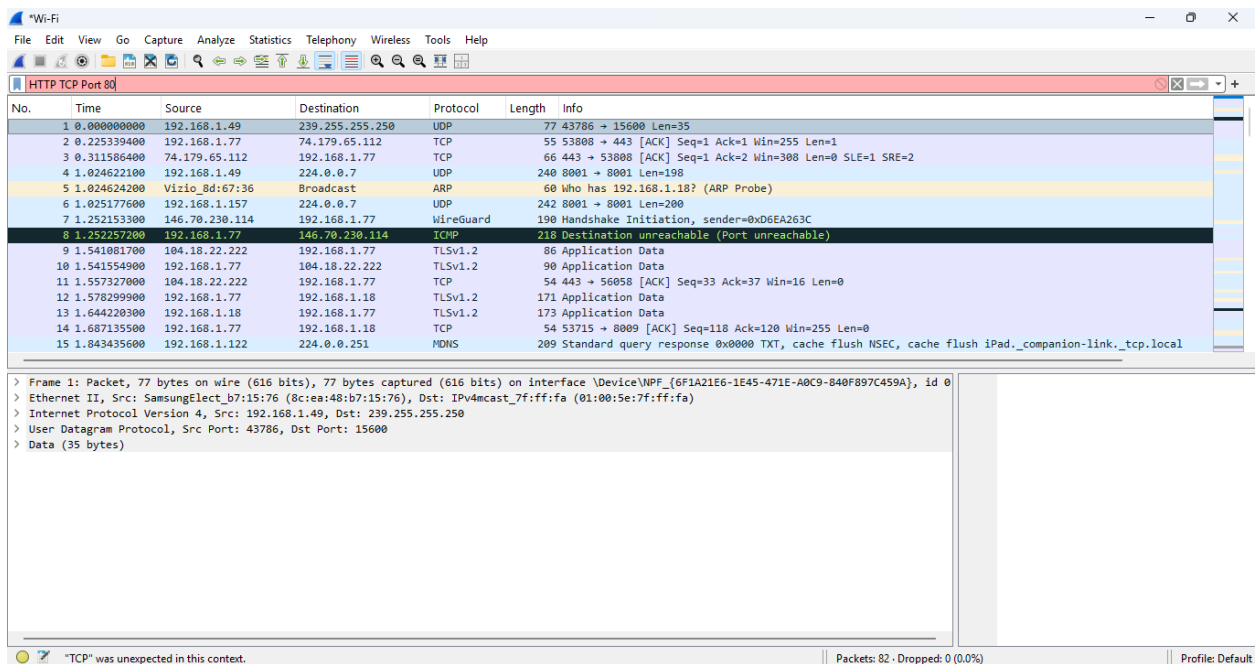
##Comment - This instruction was a bit tricky for me as I assumed that I did not have a network interface card as I use a laptop and not a desktop. After research I found that Network interface cards do exist on laptops but they're just built-in instead of being big physical cards like on a desktop. On a laptop, your Network Interface Card (NIC) will often show up as Wi-Fi (wireless) or Ethernet (if you plug in a cable). A tip to find your NIC in Wireshark is to find the one with active packet movement (the sparkline graph jumping up and down).

Step 3: Take a screenshot



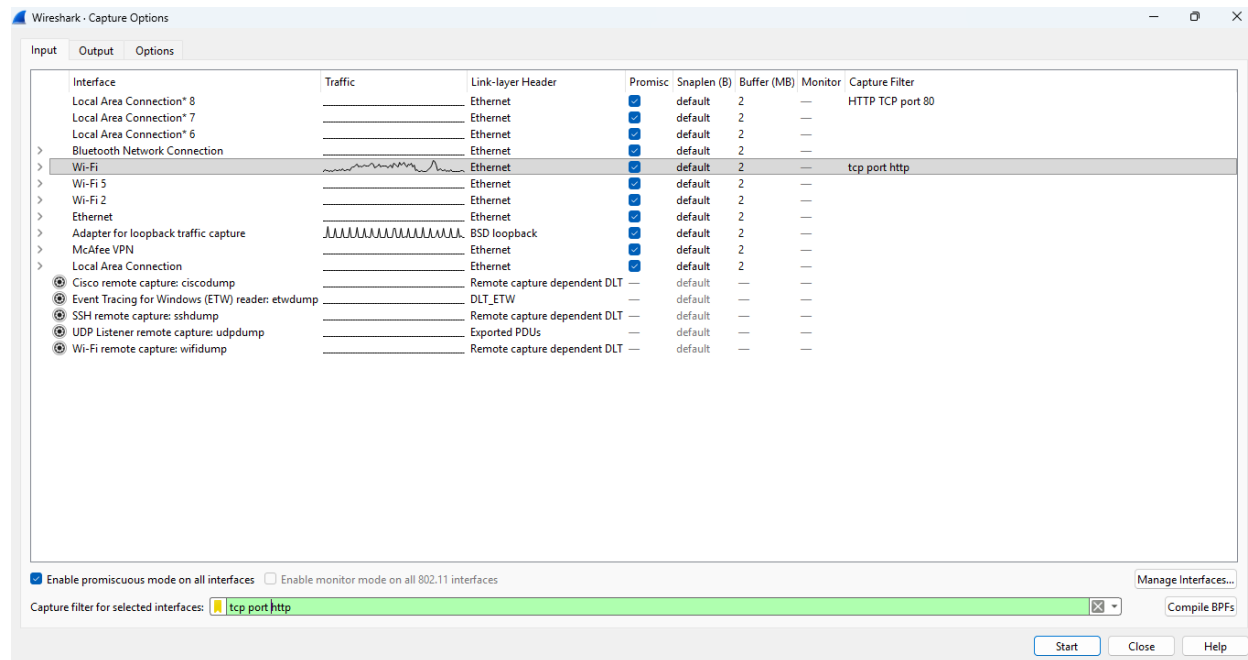
Step 4: Click stop (red) next to the first button on the toolbar of the Wireshark data capture window.

Step 5: Click the Capture drop-down menu at the top of the screen. Choose HTTP TCP port 80 from Capture Filter, then OK.

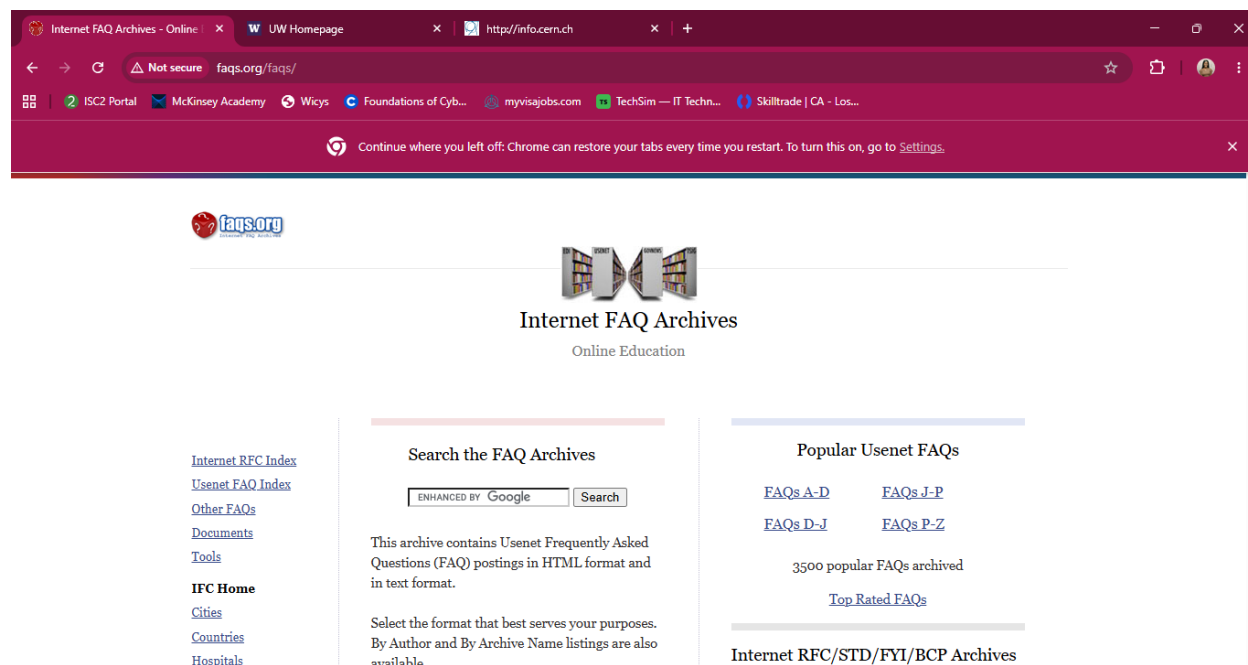


##Comment – None of my data packets captured returned a value for HTTP TCP Port 80 filter. I suspect this is because most websites have adopted using the HTTPS instead of HTTP. I realized soon after that I was using the display filter instead of the capture filter. I

then went back to the homepage to use the capture filter and used `tcp port http` as my filter.



I then visited some http websites



Step 6: Click stop (red) in the capture drop-down menu

Wireshark packet capture showing a list of network packets. The selected packet (No. 9) is a TCP Reset (RST) from 192.168.1.77 to 192.168.1.77. The info column shows "54 [TCP Dup ACK 8#1] 80 -> 60604 [ACK] Seq=1 Ack=428 Win=64128 Len=0".

Frame 1: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{6F1A21E6-1E45-471E-A0C9-840F897C459A}, id 0
 Ethernet II, Src: AzureWaveTec_18:74:f2 (9c:c7:d3:18:74:f2), Dst: AskeyCompute_e2:85:3d (90:d3:cf:e2:85:3d)
 Internet Protocol Version 4, Src: 192.168.1.77, Dst: 192.168.1.77
 Transmission Control Protocol, Src Port: 60604, Dst Port: 80, Seq: 0, Len: 0

Step 7: Scroll down until you see a line that has “GET/HTTP/1.1” in the info column

Wireshark packet capture showing a list of network packets. The selected packet (No. 7) is an HTTP GET request from 192.168.1.77 to 192.168.1.77. The info column shows "481 GET / HTTP/1.1".

Frame 7: Packet, 481 bytes on wire (3848 bits), 481 bytes captured (3848 bits) on interface \Device\NPF_{6F1A21E6-1E45-471E-A0C9-840F897C459A}, id 0
 Ethernet II, Src: AzureWaveTec_18:74:f2 (9c:c7:d3:18:74:f2), Dst: AskeyCompute_e2:85:3d (90:d3:cf:e2:85:3d)
 Internet Protocol Version 4, Src: 192.168.1.77, Dst: 192.168.1.77
 Transmission Control Protocol, Src Port: 60604, Dst Port: 80, Seq: 1, Ack: 1, Len: 427
 Hypertext Transfer Protocol

Step 8: In the bottom pane, you will see a bunch of numbers to the left (It is the packers contained in hexadecimal). Just to the right, you will see the contents of the packet in a column, Select the text: [Host:www.faqs.org](http://www.faqs.org)

Wireshark packet capture showing a TCP stream for HTTP. The packet list shows a GET request (red) and a 200 OK response (blue). The packet details pane shows the raw data of the response, which is an HTML document.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.1.77	199.231.164.68	TCP	66	60604 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2	0.002827700	192.168.1.77	199.231.164.68	TCP	66	59749 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3	0.062880300	199.231.164.68	192.168.1.77	TCP	66	80 → 60604 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1432 SACK_PERM WS=128
4	0.062258500	192.168.1.77	199.231.164.68	TCP	54	60604 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
5	0.063348900	199.231.164.68	192.168.1.77	TCP	66	80 → 59749 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1432 SACK_PERM WS=128
6	0.063499200	192.168.1.77	199.231.164.68	TCP	54	59749 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
7	0.064151200	192.168.1.77	199.231.164.68	HTTP	481	GET / HTTP/1.1
8	0.118808300	199.231.164.68	192.168.1.77	TCP	54	80 → 60604 [ACK] Seq=1 Ack=428 Win=64128 Len=0
9	0.118809400	199.231.164.68	192.168.1.77	TCP	54	[TCP Dup ACK (81)] 80 → 60604 [ACK] Seq=1 Ack=428 Win=64128 Len=0
10	0.118822900	199.231.164.68	192.168.1.77	HTTP	629	HTTP/1.1 301 Moved Permanently (text/html)
11	0.131862100	192.168.1.77	199.231.164.68	HTTP	486	GET /faq/ HTTP/1.1
12	0.215778900	199.231.164.68	192.168.1.77	TCP	1486	80 → 60604 [ACK] Seq=576 Ack=860 Win=64128 Len=1432
13	0.215772600	199.231.164.68	192.168.1.77	TCP	1486	80 → 60604 [ACK] Seq=2008 Ack=860 Win=64128 Len=1432 [TCP PDU reassembled in 14]
14	0.215773900	199.231.164.68	192.168.1.77	HTTP	67	HTTP/1.1 200 OK (text/html)
15	0.256438300	192.168.1.77	199.231.164.68	HTTP	384	GET /style/faq.css HTTP/1.1

Packet 9 details (raw data):

```

0000  90 d3 cf e2 85 3d 9c c7 d3 18 74 f2 08 00 45 00  ....t...E
0010  01 d3 af 37 40 00 80 06 1b cc c0 a8 01 4d c7 e7  ...7@....R
0020  a4 44 ec bc 00 50 7f 62 b9 ee 28 c4 1e ab 50 18  ...P-b-(-P
0030  00 ff a8 35 00 00 47 45 54 20 2f 20 48 54 54 50  ...5GET/HTTP
0040  2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 77 77 77 2c  /1.1Host: www
0050  66 61 71 73 2e 6f 72 67 0d 0a 43 6f 6e 6e 65 63  faqs.org:Conne
0060  74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65  tion: keep-alive
0070  0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75  ..Upgrade-Insecu
0080  72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d 0a  re-Request: 1
0090  55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69  User-Agent: Mozilla/5.0 (Windows
00a0  6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73  lla/5.0 (Windows
00b0  20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b  NT 10.0 ; Win64;
00c0  20 78 36 34 29 20 41 70 70 6c 65 57 65 62 4b 69  x64) AppleWebKit/
00d0  74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c  t/537.36 (KHTML,
00e0  20 6c 69 6b 65 20 47 65 63 6b 6f 29 20 43 68 72  like Gecko) Chr
00f0  6f 6d 65 2f 31 35 33 2e 30 2e 30 2e 30 20 53 61  ome/153.0.0.0 Sa
0100  66 61 72 69 2f 35 33 37 2e 33 36 0d 0a 41 63 63  fari/537.36 Acc
0110  65 70 74 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61  ept: text/html,a
0120  70 70 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c  pplicati on/xhtml

```

Step 9: Right-click the packet GET /HTTP/1.1 and select Follow, then TCP Stream to see HTTP request (Red) and HTTP response (Blue) messages as can be seen below.

Wireshark · Follow TCP Stream (tcp.stream eq 0) · Wi-Fi (tcp port http)

```

GET / HTTP/1.1
Host: www.faqs.org
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9

HTTP/1.1 301 Moved Permanently
Date: Mon, 14 Sep 2026 03:17:21 GMT
Server: Apache
X-Frame-Options: SAMEORIGIN
X-XSS-Protection: 1; mode=block
X-Content-Type-Options: nosniff
Location: http://www.faqs.org/faqs/
Content-Length: 233
Keep-Alive: timeout=10, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="http://www.faqs.org/faqs/">here</a>.</p>
</body></html>

GET /faqs/ HTTP/1.1
Host: www.faqs.org
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7

```

4 client pkts, 7 server pkts, 7 turns.

Entire conversation (7815 bytes) Show as ASCII No delta times Stream 0

Find: ☐ Case sensitive Find Next

Filter Out This Stream Print Save as... Back Close Help

Answer Questions:

1. What do the different colors mean? The red text is the HTTP GET request. This is data sent from my machine/computer to the web server. The blue text is the HTTP response. This is data sent from the web server to my computer.
2. How many packets does your computer send/receive in a single mouse click when you visit a website? There is no fixed number. There could be multiple packets for a single click. This is because a single click can lead to computer to do many things like send a request to the web server, receive the webpage data, perform a DNS lookup, establish TCP connection, download images/fonts/javascript, etc.
3. Could you organize or filter the traffic to make it easier to understand? Yes. Organizing or filtering the traffic to make it easier to understand is possible. You can do this by using display filters. This lets you organize exactly what you want to see such as data coming

from different ports (tcp port 80 or 443) or different type of traffic (DNS, HTTP). You can also click the column headers to sort the packets by time, which can allow you to see the latest packets or earliest packets captured. You can also sort by source, destination, protocol, length, info.

4. Why does your computer send so many packets? Why not send just one big packet? Networks have a maximum size on how big a packet can be. If your computer tried to send a webpage in one packet, it would be too big to fit on the network. It has to be split into smaller packets. This is more reliable because if one packet get lost or damaged, instead of sending the entire webpage, only the small piece would need to be sent again. Smaller packets also allow for security. With smaller packets, you can inspect faster and detect malicious activities easily.

5. What do SYN, ACK, FIN, and GET mean? SYN, ACK and FIN are control flags used in TCP to manage connections and traffic flow. SYN stands for synchronize, it is used to start/establish a connection between two computers. ACK stands for acknowledge, it is used to acknowledge that data or a connection request was received. FIN stands for finish, it closes a TCP connection safely when data transfer is completed. GET is an HTTP command used to request data from a web server. It basically asks a server to send a specific file, image or a webpage.

6. Why do some packets have sequence numbers (Seq= ...)? Sequence numbers are a way to basically keep track of the order of the data that is being sent. If the packets arrive out of order, TCP can use the sequence number to put them in the correct order. It also helps detect if a data packet is missing